CLASP-seq protocol

Cell pairing, targeted capture and sequencing. An assembled protocol draft with author-completion fields in red.

Working document. Red **TO COMPLETE** fields identify missing or conflicting author inputs. Complete the fields affecting a procedure before using it at the bench. Recorded settings are not a claim of experimental optimization.

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01 Scope and use

CLASP-seq combines engineered TCR-expressing Jurkat cells with HLA-peptide-expressing K562 cells. After pairing, stabilization and sorting, custom 10x capture and targeted amplification recover receptor and HLA-peptide identities for Nanopore sequencing. The documented constructs use class I HLA-peptide/B2M architecture. [S1 pp.1-3]

Recorded settings below come from the author-supplied source synthesis. They describe the documented workflow; they have not been established as universal optima. Red completion fields remain attached to the steps they affect.

TO COMPLETE - C14
Supply method authors and contact, released protocol version, approval date, persistent identifier, reuse terms and material access routes. The document-generation date is not the protocol version.

Use this document with the library-preparation, materials, controls, loading/pooling and analysis guides. Start at the assay only when both cell libraries have qualified identities, expression, viability and representation, supported by a sequence-level reference manifest.

A co-positive event is not a sequence-resolved pair. Co-recovered identities require controls and a defined analysis method before an interaction can be assessed. Keep reads, cells, sorted events, intact pairs, loaded material and recovered partitions in separate fields. [S1 pp.1-4]

02 Plan the run

Reserve the sorter, compatible Chromium instrument and reverse-transcription thermocycler for the same assay day. The source directs immediate progression from sorting to 10x; no validated overnight pause for sorted pairs is supplied. Record actual hold and transfer times. [S1 pp.1-3]

Module	Source planning estimate
Plasmid-library preparation, if needed	1-2 weeks
Stable cell-library preparation, if needed	1-2 weeks
Pairing through 10x reverse transcription	One long day
Targeted recovery PCR and QC	2-3 days
Nanopore library preparation	1-2 days
MinION sequencing	2-3 days
Analysis	TO COMPLETE: measured runtime and hardware

These are source planning estimates, not guaranteed turnaround or measured hands-on time. Use the preparation guide if starting from plasmids or sequence lists.

Qualify a small run first

1. Perform a flow pilot with the intended libraries and record the co-positive gate frequency, denominator, timing and control behavior. The source describes a 2-4 h pairing window. [S1 p.1]
2. Before scaling, use a small run through stabilization, sorting, capture, PCR and analysis to qualify downstream recovery. This end-to-end qualification is a documentation recommendation; the source does not supply its reference results.
3. Choose positive, specificity/background and process controls with the method authors. Record expected results and progression criteria before the full experiment.

TO COMPLETE - C13

Attach named, experimentally supported control materials, replicate design, reference-run ranges, post-sort composition checks and stage-specific acceptance criteria. Input-library aliquots measure representation; they do not alone establish pair-background.

03 Assign samples before handling cells

Use durable sample identifiers from the first aliquot through sequencing. The role labels below are examples, not required naming syntax. [S1 pp.1-2]

Biological sample	Capture reaction	PCR2 products
Input TCR cells	IN_TCR	IN_TCR
Sorted experimental preparation	EXP_PAIRS	EXP_TCR and EXP_HLA
Input HLA-peptide cells	IN_HLA	IN_HLA

Each of the three capture reactions produces its own PCR1 material. Split experimental PCR1 material into two PCR2 branches. The two experimental products therefore share one capture reaction; the input reactions have independent barcode namespaces. An identical barcode string in different reactions does not establish a pair.

1. Before labeling or pairing, reserve separate input TCR and input HLA-peptide aliquots. The source specifies 100 µL of each at 1,000 cells/µL. [S1 p.1]
2. Record experiment, replicate, library, channel and branch identifiers; count method; viability; collection time; and holding conditions.
3. Allocate additional controls and replicates to explicit samples, channels and indexes. Preserve the mapping in the run record.

TO COMPLETE - C07
Confirm allowable input holding conditions and duration, any input treatment required to match the experimental preparation, and the final sample/channel allocation. Reconcile individual-cell and intact-pair counting before loading.

04 Label the TCR cells

TO COMPLETE - C03
Before staining, supply the tracer product, stock molarity, solvent/reconstitution, working concentration, volume per cell number, pellet settings and maximum post-stain hold. The source's 'dilute violet' label does not define an absolute concentration.

The recorded staining sequence is below. It uses PBS during staining. The source describes a 1:50 preparation of 'dilute violet', then 1 µL of that preparation per 1 mL PBS; neither dilution resolves the missing original stock concentration. [S7 p.1]

1. Pellet up to 3 × 10^6 TCR cells, remove the medium, wash once with PBS and pellet again using the author-confirmed conditions.
2. Resuspend in the specified PBS/tracer mixture. Do not prepare the mixture until the red concentration fields above are resolved.

3. Incubate for 20 min at 37 °C.
4. Pellet and remove the staining supernatant.
5. Add 1 mL medium and incubate for 5 min at 37 °C to quench.
6. Pellet and resuspend in fresh medium. Measure recovery and inspect the fluorescence distribution with the planned controls.

Checkpoint
Record cell recovery, viability, staining settings and control distributions. Do not infer absence of dye transfer without a control that tests it.

05 Form cell pairs

The source example combines 1 × 10^6 violet-labeled TCR cells in Jurkat medium and 1 × 10^6 GFP-positive HLA-peptide cells in K562 medium in a 6-well plate. Pairing lasts 2-4 h. A platform-shaker cycle of 130 rpm for 30 s followed by 30 min rest repeats during incubation. [S1 p.1]

TO COMPLETE - C04
Specify the two medium formulations and volumes, total volume, final cell density, temperature/CO_2, plate and shaker geometry/orbit, exact sampling time and qualified scaling range. Do not infer pairing temperature from the adjacent staining or stabilization steps.

1. Confirm both input populations and record their actual counts, viability, expression and library representation.
2. Combine the populations using the author-confirmed recipe and begin the recorded mixing/incubation schedule. Retain start, mixing, sampling and finish times for each batch.
3. During the final half-hour, prepare and calibrate the sorter. The source specifies a 130 µm chip; its instrument-specific configuration must accompany the final method. [S1 p.1]
4. Carry the preparation directly into the confirmed stabilization procedure. Retain batch identity throughout subsequent handling.

Collection planning
The source plans for 20,000 collected doublets. That is a local collection target, not a validated universal requirement. Sorter events, recovered intact pairs and captured partitions are different quantities; plan from pilot measurements and record losses at each stage.

06 Stabilize with BS3

TO COMPLETE - C05
Resolve the incubation conflict before starting: the assay overview specifies 30 min at 37 °C; the dedicated BS3 sheet specifies 30 min at room temperature with rotation. Supply the selected temperature in °C, supporting run, rotation setting, exact fresh-stock preparation and glycine quench concentration/pH.

The dedicated sheet records the sequence below. It does not establish that the conflicting temperatures are interchangeable. [S1 p.1; S8 p.1]

1. Gently transfer paired cells and pellet at 300 × g for 3 min.
2. Remove medium, wash with PBS and resuspend in 95 µL PBS in a secure-lid PCR tube. The note describes approximately five moderate P200 mixes without bubbles.
3. Prepare BS3 immediately before use using the confirmed product and formulation. The note adds 35 µL PBS to a 2 mg tube and labels the result a 100 mM, 20X stock; verify this preparation before treating that nominal concentration as a recipe.
4. Immediately add 5 µL of the confirmed stock to the 95 µL cell suspension. The recorded total is 100 µL. A 100 mM stock at this dilution would give nominal 5 mM BS3; this is a calculation from the source label, not a measured concentration.
5. Incubate for 30 min at the author-confirmed temperature and rotation setting.
6. Add 11 µL of the author-defined '10X glycine quench' and rotate for 10 min at the specified room temperature.
7. Transfer to a larger tube containing 400 µL medium while awaiting sorting.

TO COMPLETE - C05
Complete the cell number per reaction, PBS formulation/pH, wash volumes, BS3 product/form and solubility/appearance criteria, absolute glycine formulation, tube/rotation setup and maximum holding interval. An undefined '10X' quench or cloudy preparation cannot serve as an acceptance criterion.

07 Sort and recover the preparation

TO COMPLETE - C06
Supply sorter model, chip/nozzle, pressure, event rate, mode, filter pore size, post-stabilization pellet settings, collection/concentration method and allowed hold/batch-combination intervals. Attach controls, FCS files, compensation/unmixing settings, annotated gates and post-sort composition/recovery criteria.

1. Process waiting tubes individually. The dedicated sheet directs pelleting, resuspension in 300 µL HBSS + 2% FBS without DAPI, filtration and immediate sorting. Keep later tubes in medium until their turn. [S8 p.1]
2. Use the violet/GFP co-positive gate with a documented hierarchy that retains the intended cell pairs and handles larger aggregates. A routine singlet-only gate would discard the intended population.
3. Collect into 1 mL Jurkat medium in a 5 mL tube. The source lists a 130 µm chip, 'sort speed 2' and an example density of 2 × 10^6 cells per 300 µL. These settings require the completed C06 instrument and operating records. [S1 p.1]
4. Record acquired events, parent-gate denominators, co-positive events and collected events separately. Measure recovered cell counts and assess intact-pair composition using the confirmed method.
5. Proceed immediately to the defined 10x preparation. If combining batches of the same libraries, retain each contribution and its handling times; the source permits combining batches only when timing is similar.

Co-positive events backgating into a doublet scatter region do not prove that every event contains exactly one TCR cell and one HLA-peptide cell. Preserve evidence for aggregates, dissociation and loss rather than reporting every collected event as an intact pair.

08 Perform custom 10x capture and RT

Use the source-specified Chromium GEM-X Single Cell 3′ v4 chemistry and the archived CG000731 Rev B guide, with the CLASP-specific supplement completed below. This is a custom capture workflow, not a standard 5′/V(D)J library. Confirm compatibility with the receiving laboratory's Chromium instrument. [S1 pp.1-2]

TO COMPLETE - C07
Release a validated loading worksheet and counting method. The source asks for 1,000 cells/µL, at least 20 µL available on ice, and also says 'load 20K doublets'. Resolve cells versus intact pairs; define loaded material and target recovered partitions separately. Do not transfer a pair count into a standard single-cell chart without the validated conversion.

Stage	Recorded customization or requirement
RT master mix	Replace the TSO reagent volume with water; the source says the relevant sequence is built into the transcripts. TO COMPLETE: exact reagent, volume and supported construct architecture.
Chip/GEM generation	Follow the pinned vendor steps, including unused-channel handling. TO COMPLETE: channel map, instrument configuration and loading worksheet.
GEM transfer/RT	Transfer to PCR tubes on ice and perform RT under the pinned program. TO COMPLETE: full program and allowed holds.
Post-RT cleanup	Elute in 25 µL for targeted PCR1. TO COMPLETE: exact cleanup stage, reagents, volumes and validated adaptation.

Keep unchanged manufacturer steps and custom modifications distinguishable in the bench copy. Record kit/chip lots, instrument/firmware, raw counts, calculations, timestamps and deviations. The custom changes require assay validation; the manufacturer guide alone does not validate them.

09 Recover targets by PCR1 and PCR2

PCR1: three separate capture products

The assay overview assigns CZP47 + CZP48 to each of IN_TCR, EXP_PAIRS and IN_HLA. It refers to the 10x SPRI cDNA-cleanup procedure after PCR1. [S1 p.2]

TO COMPLETE - C09
Supply a complete PCR1 mix for cleaned 10x cDNA: template allocation, reaction volume, every stock/final concentration, enzyme, full thermocycler program, cleanup and verified pause points. The separate worksheet is headed 'STARTING VIRUS'; its unitless entries and '11 ROUNDS, 52 ANNEALING' note do not define the 10x recipe. Keep virus-input and 10x-input recipes separate.

PCR2: four target-recovery branches

Split the experimental PCR1 material between TCR and HLA-peptide recovery. The overview records the following assignments; the red reconciliation fields must be closed before ordering or using primers. [S1 p.2; S10 p.1]

Branch	Recorded forward / reverse	Required confirmation
IN_TCR	CZP32UNI / CZP48.1	TO COMPLETE: construct match; .1 versus .01 alias.
EXP_TCR	CZP32UNI / CZP48.2	TO COMPLETE: construct match; .2 versus .02 alias.
EXP_HLA	CZP31 / CZP48.3	TO COMPLETE: CZP31 is obsolete for the current HLA construct; reconcile CZP31.0 and .3/.03.
IN_HLA	CZP31 / CZP48.4	TO COMPLETE: reconcile CZP31.0 and .4/.04 before use.

TO COMPLETE - C08
Do not use obsolete CZP31 as the current HLA recovery primer. Confirm the replacement against each released construct. Map indexed-primer aliases and establish when CZP32UNI or the registry's TCRAFT-specific CZP160 is compatible. The materials guide preserves primer identities for this review.

TO COMPLETE - C09
Supply PCR2 composition, template amounts and split, primer stocks/final concentrations, complete cycling, controls and permitted holds for each supported construct. TCRAFT construction/representation PCR is not a substitute for this missing recovery recipe.

Keep branch indexes separate from 10x partition barcodes. The index sequence alone does not establish whether or where UMIs are retained. Preserve an annotated construct/read map and sequence-confirmed reference manifest for analysis.

10 Clean up and check each product

TO COMPLETE - C10
Specify the exact cleanup product/procedure, gel/extraction conditions, post-extraction bead calculation, fragment-analysis assay and construct-specific expected sizes and acceptance criteria. The source examples near 2.2 kb for TCR and 1.6 kb for HLA-peptide are not standards for every library.

1. For the standard recorded PCR2 cleanup, combine 30 µL PCR2 product with 20 µL water, then add 30 µL SPRI beads to the resulting 50 µL sample: the stated ratio is 0.6X. Elute in 22 µL water. Complete the exact bead product, wash and handling procedure before use. [S1 p.2]

2. For the relevant TCRAFT-derived construct only, the source describes an approximately 1 kb dead-end product. If that documented product is present, the recorded branch defers initial SPRI cleanup, gel-isolates the 1.6-2.2 kb region and then performs the specified cleanup with 22 µL elution. This window is not universal to TCR constructs. [S1 pp.2-3]
3. Keep IN_TCR, EXP_TCR, EXP_HLA and IN_HLA separate through identity and quantity checks. The source uses a 1:5 dilution for TapeStation, with 1 µL diluted sample or ladder in 10 µL sample buffer with dye. Confirm the exact assay and range before using this loading format. [S1 p.2]
4. Retain complete fragment traces and quantification files. Record target size, concentration, units, dilution correction, off-target peaks and any gel selection.

Checkpoint
Confirm product identity and usable target concentration before pooling. A band or total DNA concentration alone does not establish the amount of correctly indexed target amplicon. Preserve blank/control results and any failed branch.

11 Pool and prepare the sequencing library

The source suggests 0.1 IN_TCR : 0.4 EXP_TCR : 0.4 EXP_HLA : 0.1 IN_HLA. Its calculation basis is not stated. These proportions cannot be called molar, mass or volume fractions until confirmed. [S1 p.3]

TO COMPLETE - C10
Supply the pooling basis and calculator, a worked example, total pool-volume limit, per-branch limits and validated library-preparation input. Resolve the source's 'no sample can contribute more than 19 µL' note and its 200 fmol input reference; do not turn local higher-loading observations into an optimized amount.

1. Use the loading/pooling worksheet to record each branch's target size, measured concentration and units, dilution correction, intended contribution and actual pipetted volume.
2. Prepare the pool only after the calculation basis and limits are confirmed. Keep total input to library preparation separate from the amount eventually loaded onto the flow cell.
3. Use the archived Ligation sequencing amplicons V14 (SQK-LSK114) method and compatible flow cell. The source sequence is end preparation, cleanup, adapter ligation and post-ligation cleanup, with 12 µL final elution. [S1 p.3]
4. Use stage-appropriate manufacturer buffers and retain required QC. The source's repeated 'polishing' label does not define post-ligation cleanup, and its informal suggestion to omit QC is not an acceptance criterion.

TO COMPLETE - C11
Pin the ONT document revision, platform/flow cell, input/loading amounts, full library-preparation steps and QC. Identify each CLASP-specific change and its supporting reference run. The supplied notes do not contain this complete executable recipe.

12 Sequence, archive and assess pair evidence

1. Perform the selected flow-cell check and retain its report. Use the released platform and assay criteria for proceeding; local pore-count preferences in the notes are not validated assay-sensitivity thresholds.

2. Record run ID, instrument, flow cell and kit lot, MinKNOW version, basecalling software/model and settings. The source enables POD5 output and uses a 2-3 day run. Duration alone does not establish adequate sampling of each branch. [S1 p.3]

3. Retain raw signal, reads, sample/index maps, acquisition reports and analysis configuration. Verify filenames, counts, sizes and checksums against the acquisition copy when archiving.

4. Use the assay-specific analysis to assign branches, recover partition identities, match constructs and evaluate pair evidence. Preserve ambiguous/unassigned records, exclusions, input representation and control results.

TO COMPLETE - C11
Supply acquisition/basecalling configuration, branch coverage requirements, run stopping criteria and validated archive/retention requirements.

TO COMPLETE - C12
Release the complete read architecture, demultiplexing/barcode rules, reference manifest, assignment and ambiguity policy, background/normalization/calling method, code/environment and a worked example with expected outputs. An uncaptioned result image is not a reproducibility target.

Interpret associations within the experimental sample/channel. Do not pair identities across the independent input channels, label PCR reads as independent biological observations, or report an unobserved combination as a verified nonbinder. Keep measured evidence and downstream model scores in separate fields.

Complete the experiment record
Retain the protocol and manifest versions, reagents and instruments, actual timing, count units, sorting gates, capture allocation, primer/index map, PCR and fragment QC, pooling calculation, raw-data checksums, controls, exclusions and analysis revision. The companion worksheet provides blank fields for these records.

Source notes

Sources are the method authors' laboratory records, as transcribed in the supplied consolidation. The internal source files are not included in this download.

S1. Cell_x_Cell TCR_HLA-pep Pairing Assay V1. Author-supplied laboratory notes, pp.1-4; internal workflow label v20260501. Statements and page references transcribed from the supplied source-checked synthesis dated September 11, 2026; underlying PDFs were not re-reviewed for this document.

S7. Violet cell tracer staining. Author-supplied laboratory notes, p.1; staining sequence and incomplete stock specification, as recorded in the supplied synthesis.

S8. BS3 Crosslinking. Author-supplied laboratory notes, p.1; stabilization and pre-sort handling, including the temperature conflict, as recorded in the supplied synthesis.

S9. PCR1, PCR2 for Cell_x_Cell Assay - STARTING VIRUS, 1st ROUND PCR. Author-supplied worksheet, p.1; incomplete virus-input PCR record, not a complete 10x PCR recipe, as recorded in the supplied synthesis.

S10. CZ PRIMERS AND GENEBLOCKS - CZP1-. Author-supplied oligo registry, pp.1-2; primer identities, sample indexes and deprecation notes, as recorded in the supplied synthesis.

Protocol and resources